Python Collections

List

List

- A **list** is a collection which is:
  - **Ordered** (items have a defined order and can be accessed using indexes).
  - **Changeable (Mutable)** (we can add, update, or remove items).
  - **Duplicate values** are allowed in the list.
- Lists are written inside **square brackets [ ]**.

Creating a List

```
mylist1 = [10, 20, 30, 40, 50]
mylist2 = ["apple", "banana", "cherry"]
mylist3 = ["apple", 10, "banana", 20]  # mixed data types
mylist4 = list()   # empty list

print(mylist1)   # [10, 20, 30, 40, 50]
print(mylist2)   # ['apple', 'banana', 'cherry']
```

Accessing List Items

- Indexing starts from **0**.
- Negative indexing starts from **-1** (last element).

```
mylist = ["apple", "banana", "cherry"]
print(mylist[0])    # apple
print(mylist[-1])   # cherry
```

Range of Indexes

```
mylist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
print(mylist[2:5])    # ['cherry', 'orange', 'kiwi']
print(mylist[-4:-1])  # ['orange', 'kiwi', 'melon']
```

Changing Item Value

```
mylist = ["apple", "banana", "cherry"]
mylist[1] = "orange"
print(mylist)  # ['apple', 'orange', 'cherry']
```

Loop Through a List

```
mylist = ["apple", "banana", "cherry"]
for item in mylist:
    print(item)
```

Check if Item Exists

```
mylist = ["apple", "banana", "cherry"]
if "apple" in mylist:
    print("Yes, apple is present")
```

List Functions

```
mylist = ["apple", "banana", "cherry"]
print(len(mylist))          # 3 (length of list)
print(mylist.count("apple"))  # count occurrences
```

Sorting & Reversing

```
mylist = ["cherry", "mango", "banana", "apple"]
mylist.sort()
print(mylist)  # ['apple', 'banana', 'cherry', 'mango']

mylist.sort(reverse=True)
print(mylist)  # ['mango', 'cherry', 'banana', 'apple']

mylist.reverse()
print(mylist)  # ['apple', 'banana', 'cherry', 'mango']
```

Note: You cannot sort heterogenous data.

```
mylist=["a", 1, "x"]
mylist.sort() #TypeError
print(mylist)
```

Adding Items

```
mylist = ["apple", "banana"]
mylist.append("cherry")        # add at end
mylist.insert(1, "orange")     # insert at specific position
print(mylist)  # ['apple', 'orange', 'banana', 'cherry']
```

Removing Items

```
mylist = ["apple", "banana", "cherry"]
mylist.remove("banana")   # remove by value
mylist.pop(0)             # remove by index
del mylist[1]             # delete specific item
mylist.clear()            # clears all items
```

Copying a List

```
mylist1 = ["apple", "banana"]
mylist2 = mylist1.copy()
print(mylist2)  # ['apple', 'banana']
```

Joining Lists

```
list1 = ["a", "b", "c"]
list2 = [1, 2, 3]
# Approach 1: +
print(list1 + list2)  # ['a', 'b', 'c', 1, 2, 3]
# Approach 2: extend()
list1.extend(list2)
print(list1)  # ['a', 'b', 'c', 1, 2, 3]
```

Key Point: Lists are **mutable** → you can **modify** them anytime.